Biliary Atresia, Part Twelve: Liver Transplantation. Indications and Timing, Split Versus Living Donor Grafts, and the Anatomical Surprises of the Splenic Malformation.

Imagine a seven-month-old boy who had a Kasai operation for biliary atresia at fifty-two days of age. The operation never really worked. Three months after surgery his bilirubin was nine milligrams per decilitre, and now it is fourteen. His weight has fallen to a z-score of minus two point four, which means he is well below the second centile. There is fluid in his abdomen. An ultrasound done for the transplant team shows several small spleens, and the radiologist mentions that the portal vein runs in front of the duodenum rather than behind it. His mother says that she wants to donate part of her own liver. The questions for the clinician are simple to state and hard to answer well. When should this child be referred and listed for transplantation? Which kind of graft should he receive? And what does his unusual anatomy mean for the operation and for his chances afterwards?

The main idea of this chapter is that liver transplantation is the second operation in the plan for most children with biliary atresia, not a sign that the first operation failed in some shameful way. The decisions that change the outcome are made months before the transplant itself. They are decisions about when to refer, which graft to use, and how well the surgical team understands the child's blood vessels before the first incision. A child who is referred early, fed well, vaccinated fully, and operated on with a carefully chosen graft by a team that has already mapped the veins does very well. A child who is referred late, after the third episode of cholangitis and with wasted muscles, does much worse.

A few terms first. The Kasai operation, also called a hepatic portoenterostomy, removes the scarred remnant of the bile ducts and joins a loop of small bowel directly to the cut surface of the liver at the porta hepatis, so that bile can drain from tiny surviving ductules. Biliary cirrhosis is the scarring of the whole liver that follows when bile does not drain. A graft is the piece of liver that is transplanted. The left lateral segment is the part of the liver to the left of the falciform ligament, made of segments two and three, and it is the graft that fits most infants. A split graft is made by dividing one deceased donor liver into two parts, usually a left lateral segment for a child and the remaining larger part for an adult. A living donor graft is a left lateral segment taken from a living adult, usually a parent. The Pediatric End-Stage Liver Disease score, shortened to PELD, is the number used in the United States to rank children under twelve on the waiting list.

The evidence in this chapter comes from several sources. The main review of indications and timing is by Sundaram, Mack, Feldman and Sokol, published in Liver Transplantation in 2017. Waiting list and post-transplant outcomes come from the Studies of Pediatric Liver Transplantation registry, known as SPLIT, reported by Utterson in 2005, by Anderson in 2008 and by Ebel in 2020. Long-term results of living donor transplantation come from the Japanese Liver Transplantation Society registry of two thousand and eighty-five children, reported by Kasahara and colleagues in 2018. The comparison of living donor and split grafts comes from a United States national database analysis by Dalzell and colleagues in 2022. The question of whether a prior Kasai harms the transplant comes from cohorts reported by Dong in 2025 and by Lemoine in 2022. Guidance comes from the 2026 practice guideline on evaluation of the paediatric liver transplant candidate from the American Association for the Study of Liver Diseases, the American Society of Transplantation and the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, and, for comparison, from the 2017 joint North

American and European guideline on the cholestatic infant and the 2016 European Association for the Study of the Liver guideline on liver transplantation.

Part one. Five roads from a Kasai to a graft

Why does a child with biliary atresia end up needing a new liver? There are two very different populations, and it helps to keep them apart.

The first population is the infants whose Kasai never drained. If bile does not flow by about three months after the operation, the liver continues to scar quickly, and cirrhosis with all its consequences develops within months. These children usually need a transplant before their second birthday. About half of all children with biliary atresia need a liver in the first two years of life, and almost all of them belong to this group.

The second population is the children whose jaundice did clear. Their livers drain, but the scarring that began before the Kasai, and sometimes continues slowly afterwards, leads to cirrhosis over years. These children reach transplantation later, sometimes in adolescence or adulthood, by one of four further roads.

The first of these roads is cholangitis, which is bacterial infection of the bile ducts. It is very common in the first two years after a Kasai: forty to eighty percent of children have at least one episode before the age of two, and a quarter have several. Repeated cholangitis carries roughly three times the risk of early failure of the Kasai. The second road is portal hypertension, meaning high pressure in the portal vein because blood cannot pass easily through the scarred liver. It causes varices that bleed, fluid in the abdomen, called ascites, and an enlarged overactive spleen, called hypersplenism. The third road runs through the lungs. Hepatopulmonary syndrome is a condition in which the small vessels of the lung dilate, so blood passes through them without picking up enough oxygen. Portopulmonary hypertension is the opposite problem: the lung arteries narrow and the pressure in them rises. Each of these sets its own clock for transplantation, and both were the subject of an earlier chapter. The fourth road is growth. Some children simply cannot be fed well enough to grow and to keep their muscle, even with tube feeding. Rarely, a liver tumour appears, and hepatocellular carcinoma develops in about one percent of children with biliary atresia.

Part two. Prognosis sets the date

If transplantation is likely, how do we know when it is coming? The most useful single number is the total bilirubin three months after the Kasai.

Data from the North American Childhood Liver Disease Research Network, summarised in the 2017 review, show the size of this effect. Among children whose bilirubin had fallen below two milligrams per decilitre three months after surgery, eighty-four percent were alive with their own liver two years later. Among children whose bilirubin was still above six milligrams per decilitre, only sixteen percent were. That is a difference between most children and one child in six, and it was highly statistically significant. Looking further ahead, children whose jaundice clears by three months have a ten-year survival with their native liver of about seventy-five to ninety percent, whereas only about twenty percent of children with persistent jaundice survive three years without a transplant.

The practical lesson is that the three-month bilirubin is the most important date in the life of a child after Kasai. If the bilirubin is above six milligrams per decilitre at that point, or is rising, the question is

no longer whether this child will need a liver, but when. The review recommends that a child without good bile flow by three months should be evaluated for transplantation by about six to nine months of age. In practice that means the referral letter should go out the same week, not after the next complication.

Part three. When to call the transplant team

What are the triggers for referral? It helps to think of them as a list that a general hepatologist or paediatrician can keep in mind at every clinic visit.

The first trigger is a failed Kasai, defined by a bilirubin above six milligrams per decilitre at three months or a bilirubin that keeps rising. This calls for immediate referral. The second is growth failure, meaning a weight or height z-score below minus two despite tube feeding. This matters not only for the child's quality of life, but because poor growth predicts death and graft loss after the transplant, as we will see. The third is recurrent or resistant cholangitis. The fourth is portal hypertension that cannot be controlled, such as recurrent variceal bleeding despite band ligation, ascites that does not respond to diuretics, or severe hypersplenism. A single variceal bleed that is controlled by banding is not, on its own, always an indication. The fifth trigger is lung disease: hepatopulmonary syndrome is usually an indication for transplant, and portopulmonary hypertension requires treatment before the operation, because a mean pulmonary artery pressure above fifty millimetres of mercury is a contraindication at many centres. The sixth trigger is synthetic failure, meaning a prolonged clotting time that does not correct with vitamin K, or encephalopathy. This is a late sign in children, and we should not wait for it. Finally, intractable itching, poor quality of life and liver tumours are all reasons to refer.

The 2026 guideline from the American liver, transplant and paediatric gastroenterology societies makes earlier referral its central message. It advises referral of children with chronic liver disease when the first complication appears, whether that is growth failure, recurrent cholangitis, severe itching, portal hypertension, encephalopathy or coagulopathy that cannot be corrected. The rule to remember is to refer at the first complication, not the third.

Part four. Infants die waiting

Why is timing so important? Because infants with biliary atresia are the most vulnerable people on the waiting list.

The SPLIT registry studied seven hundred and fifty-five children with biliary atresia who were listed for transplantation, most of them infants. Being an infant and having a PELD score of twenty or more were the significant risk factors for dying on the list. Older data summarised in the 2017 review put waiting list mortality at about twenty-five percent for infants under twelve months, compared with about ten percent for older children. Among the five hundred and sixty-seven children in SPLIT who did reach a transplant, three-year survival was eighty-eight percent. Infection was the most common cause of death after the transplant, accounting for thirty-seven percent of deaths, and a height or weight below minus two standard deviations predicted both death and graft loss.

The practical point is that the youngest and sickest child has the shortest runway. Centres keep infants off the mortality curve by listing early, by asking for exception points when the score understates the illness, and by evaluating a living donor at the same time as the child is listed.

Part five. Why the PELD score understates biliary atresia

The PELD score was designed to predict death on the waiting list from a few laboratory values and growth. The original version used bilirubin, the international normalised ratio, albumin, age and a fixed penalty for growth failure. The difficulty in biliary atresia is that the international normalised ratio and the albumin are often nearly normal until very late, because the problem is drainage of bile rather than loss of liver cell function. Weight is inflated by a large liver, a large spleen and ascites, so a wasted child may not register as growth-failed on the scale. And the score does not see cholangitis, itching, portal hypertension or muscle loss at all. In older United States data, the median PELD at transplant was about fifteen, and a survival benefit from transplantation appeared only above a score of about seventeen.

The United States allocation system changed in 2023. The new score, called PELD creatinine, adds creatinine as a measure of kidney function, refits the other coefficients, and treats age and growth as continuous measures rather than categories, using the lower of the height or weight z-score. At the same time, the old requirement that a child with chronic liver disease have a calculated PELD above twenty-five to qualify for the higher urgency category, called status one B, was removed, while the requirement for a serious complication such as mechanical ventilation, major bleeding or kidney replacement therapy was kept. The 2026 guideline endorses PELD creatinine and the adult equivalent, MELD three point zero, for initial priority, but it deliberately keeps the option of non-standard exception requests for children whose score does not reflect their real risk. That description fits many infants with biliary atresia. The transplant hepatologist should not hesitate to write the exception request when cholangitis, itching or growth are driving the danger.

Part six. Should every child have a Kasai first?

Some surgeons have asked whether infants with biliary atresia should go straight to transplantation, skipping the Kasai, to avoid adhesions and a Roux loop that complicate the later operation. Two recent cohorts address this.

The larger study, by Dong and colleagues from Beijing in 2025, included four hundred and seventy-five children transplanted for biliary atresia, most with living donor grafts. One hundred and fifty-one had no previous Kasai, two hundred and eighty-seven had a Kasai before ninety days of age, and thirty-seven had a Kasai after ninety days. Biliary complications after transplant were more common in the children who had a Kasai: two percent with no Kasai, six point six percent with an early Kasai and ten point eight percent with a late Kasai. Infections were also more frequent, at about eighteen percent without a Kasai and about thirty percent with one. But graft loss was almost identical, at four point six percent without and four point five percent with an early Kasai, and on multivariable analysis the Kasai itself was not an independent predictor of graft loss. Only the small late-Kasai group did worse, with graft loss of thirteen point five percent. A smaller Chicago series by Lemoine and colleagues in 2022 found three-year patient survival of ninety-two point nine percent after primary transplant and ninety point seven percent after a Kasai, a difference that was not significant.

The verdict is that a Kasai adds technical work and some infections, but it does not lower graft or patient survival, and it spares a transplant entirely in the children whose jaundice clears. The 2026 guideline recommends the Kasai as initial management, and reserves primary transplant evaluation for the infant who presents late with decompensated cirrhosis.

Part seven. Four ways to fit an adult liver into an infant

An infant's abdomen is small, and a whole liver from a child of matching size is rarely available. So most infants with biliary atresia receive what surgeons call a technical variant graft, meaning a part of a larger liver.

The first option is a whole liver from a small paediatric donor. Such donors are scarce, waits are long, the vessels are tiny, and, as we will see, the risk of hepatic artery thrombosis is highest. The second option is a reduced-size graft, in which an adult liver is trimmed down to its left side and the rest is discarded. This has largely been replaced by splitting. The third option is a split graft: one deceased adult liver is divided into two, the child receives the left lateral segment and an adult receives the rest. Splitting turns one donor into two recipients, but it is underused; only about ten percent of eligible donor livers in the United States are split. The fourth option is a living donor graft: a parent or other adult donates the left lateral segment. This allows elective timing, very short cold ischaemia time, which is the time the liver spends on ice outside the body, and careful planning. Blood group incompatible living donation is feasible in infants, whose immune systems have not yet made high levels of the antibodies against other blood groups.

Graft choice is therefore a timing decision as much as a surgical one. A living donor turns an unpredictable wait into a planned date, so the operation can happen before cholangitis, bleeding or muscle loss make the child a worse candidate. The donor's own risk is real, and it must be weighed by a separate team whose only job is to protect the donor.

Part eight. Living donor versus split grafts

How do living donor and split grafts compare? The best recent data come from the analysis by Dalzell and colleagues of the United States national transplant database for the years twenty ten to twenty nineteen. They studied nine hundred and eleven children who received a left lateral segment, five hundred and eight from living donors and four hundred and three from split deceased donor livers.

In the children whose diagnosis was biliary atresia, the difference was clear. Five-year graft survival was ninety-four percent with a living donor graft and eighty-one percent with a split graft, a significant difference. Five-year patient survival was ninety-seven percent and ninety-two percent, a difference that came close to, but did not reach, statistical significance. The same pattern appeared in all children weighing ten kilograms or less, with five-year graft survival of eighty-eight percent with living donors and seventy-six percent with split grafts.

There are important caveats. This was an unadjusted comparison without hazard ratios, and donor quality, centre experience and urgency at transplant differ between the groups. A child who receives a split graft may have been sicker or waited longer. But the direction of the effect is plausible: a living donor graft comes from a healthy, carefully selected adult, spends very little time on ice, and is transplanted on a planned day. The practical message is that split and living donor grafts are complements, not rivals. Every infant with biliary atresia should have access to both, and the 2026 guideline explicitly calls for equitable access to technical variant grafts.

Part nine. Twenty years after a living donor graft

What happens over the long term? The Japanese registry, where living donor transplantation has been the main route for decades, gives the longest view. Kasahara and colleagues described two thousand and eighty-five children who received a living donor graft for biliary atresia between nineteen eighty-nine and twenty fifteen. Their mean age at transplant was almost six years, which reflects the

many older children with cleared jaundice in the series.

Graft survival was ninety and a half percent at one year, ninety point four percent at five years, eighty-four point six percent at ten years, eighty-two percent at fifteen years and seventy-nine point nine percent at twenty years. In other words, after the first year the curve becomes shallow, and four out of five grafts are still working twenty years later. The factors that predicted graft survival were the donor's body mass index, blood group incompatibility, graft type, the recipient's age, the centre's experience and the era of transplant. Recipients aged twelve to seventeen years had the worst long-term graft survival, worse than both younger children and adults. The authors concluded that living donor transplantation is safe and effective for biliary atresia, that early referral improves short-term outcome, and that the long-term causes of graft loss, especially in adolescents, need further study.

The practical point is that the losses that remain after the first year cluster in adolescence, where non-adherence to immunosuppression, late rejection and the move to adult care all happen at once. Planning for transition begins at the transplant, not at the age of seventeen.

Part ten. Hepatic artery thrombosis

The most feared early complication of paediatric liver transplantation is clotting of the hepatic artery. The SPLIT registry analysed three thousand eight hundred and one first transplants in children between nineteen ninety-five and twenty sixteen, as reported by Ebel and colleagues in 2020.

Hepatic artery thrombosis developed within ninety days in seven point four percent of children. The risk was highest in infants under one year of age. When it happened, the consequences were severe: about half of these children lost the graft within ninety days, with an adjusted hazard ratio for graft failure of about twelve, and the adjusted hazard ratio for death within ninety days was about six. Among children who needed a second transplant, about one in five developed thrombosis again.

The counter-intuitive finding was that technical variant grafts, meaning split, reduced or living donor grafts, carried lower odds of thrombosis than whole grafts, with an odds ratio of about zero point six. The likely reason is that a left lateral segment comes with an adult-sized arterial stump, which can be joined to the infant's artery more easily, often under an operating microscope. A tiny whole liver from a tiny donor has a tiny artery. This is another argument for technical variant grafts in infants. In the first week after transplant, daily Doppler ultrasound of the hepatic artery and a low threshold for going back to the operating room are standard practice.

Part eleven. The splenic malformation rewrites the operation

We return now to the anatomy of the child in our opening case. About ten percent of infants with biliary atresia in Europe and North America have what is called the biliary atresia splenic malformation syndrome, a pattern first defined in a series from King's College Hospital in London in which twenty-seven of three hundred and eight infants were affected. In East and South Asia the syndrome is much rarer, found in fewer than three percent of infants. It is thought to reflect a disorder of the left and right patterning of the embryo.

The syndrome brings a group of anatomical anomalies, and each changes the transplant operation. The first is a preduodenal portal vein, meaning a portal vein that runs in front of the duodenum rather than behind it. It can be injured during the Kasai or the hepatectomy, and it is often small. The second is an absent infrahepatic inferior vena cava, in which blood from the lower body returns to the heart through

the azygos vein, and the hepatic veins enter the right atrium directly. The surgeon cannot use the standard technique of sewing the new liver onto the existing vena cava, and must plan a different route for the outflow of blood from the graft. The third group of anomalies includes multiple small spleens, called polysplenia, or no spleen at all, called asplenia, together with situs inversus, in which the abdominal organs are mirror-imaged, and malrotation of the bowel. The dissection may need to be performed as a mirror image of the usual operation. The fourth is congenital heart disease, which drives much of the extra mortality in this syndrome, so every such child needs an echocardiogram before listing. The fifth is susceptibility to infection, because a child without a working spleen is at risk from encapsulated bacteria, so vaccines and, where indicated, antibiotic prophylaxis matter.

None of these anomalies is a contraindication to transplantation at an experienced centre. But each one must be known before the first incision, because the plan for venous outflow and portal inflow is completely different from a standard operation.

Part twelve. What complex vessels cost

How much do these anomalies affect outcome? The historical answer is sobering. In 1974, Lilly and Starzl reported twenty-nine children transplanted for biliary atresia in the earliest era of the operation. Three had a combination of an absent inferior vena cava, a preduodenal portal vein and an abnormal origin of the hepatic artery, and all three died within days of surgery. The authors concluded that such children were highly questionable candidates.

The modern answer comes from the SPLIT registry, reported by Anderson and colleagues in 2008. They studied nine hundred and forty-four children transplanted for biliary atresia over eleven years, of whom sixty-three, or six point seven percent, had complex vascular anomalies. One-year patient survival was eighty-three percent in children with complex anomalies and ninety-three percent in those without, a ten-point gap. Graft survival, however, did not differ. The children with anomalies had more reoperations, more biliary complications including leaks, and more bowel perforations. Infection was the most common cause of death, and deaths from bacterial infection were more frequent in the group with anomalies. Two features before transplant predicted death: thrombosis of the portal vein and a preduodenal portal vein.

So, between 1974 and 2008, this anatomy went from almost uniformly fatal to a survival penalty of about ten percentage points. Most of the excess deaths now come from infection after reoperation rather than from failure on the operating table, which suggests that careful planning, meticulous technique and aggressive treatment of infection are where further gains lie.

Part thirteen. Image the veins before the knife

What should the pre-operative work-up include? A practical approach asks six questions.

First, what is the size and direction of flow in the portal vein? A Doppler ultrasound and computed tomography or magnetic resonance angiography answer this. A small or hypoplastic portal vein, or flow going the wrong way, changes the inflow plan, and the surgeon may need an interposition vein graft or a patch. Second, is the portal vein clotted? Portal vein thrombosis predicted death in the SPLIT registry, and the surgeon may need to bring blood to the graft through a jump graft from the superior mesenteric vein. Third, where are the vena cava and the hepatic veins? Venous imaging shows whether the infrahepatic vena cava is absent and how the hepatic veins enter the heart. Fourth, what are the situs, the spleen and the bowel rotation? Imaging together with the operative note from the Kasai shows

where the Roux loop lies and whether the dissection will be mirror-imaged. Fifth, what about the heart and lungs? An echocardiogram looks for cardiac defects and estimates pulmonary pressure, and pulse oximetry measured both upright and lying down screens for hepatopulmonary syndrome, as the 2026 guideline recommends. Sixth, how well do the kidneys work? Creatinine and cystatin C, combined in the equation known as the Chronic Kidney Disease in Children under twenty-five equation, give the baseline before calcineurin inhibitors, and creatinine is now part of the PELD score.

Part fourteen. The months before the graft are treatment

While the child waits, five tasks fill the time. The first is feeding. Infants with cholestasis need about one hundred and thirty to one hundred and fifty percent of normal energy intake, a formula rich in medium-chain fats, often nasogastric tube feeding, and protein that is not restricted. The second is vaccination. Live vaccines cannot be given after transplantation because of immunosuppression, so the schedule should be accelerated, and the 2026 guideline advises starting measles, mumps and rubella and varicella vaccines from six months of age in transplant candidates. This matters because up to seventy-one percent of paediatric organ recipients arrive at transplant with incomplete immunisations. The third task is screening and treating complications: oximetry and echocardiography, treatment of cholangitis, band ligation of varices, and monitoring of kidney function. The fourth task is the donor. A living donor work-up, starting with blood group and imaging of the donor liver, should begin in parallel with listing, not after a crisis. The fifth is preparation of the family: social work, dental review, teaching, and a plan for adherence to medicines from the first day.

In SPLIT, growth below minus two standard deviations predicted death and graft loss after transplant. Every month on the list is either a chance to repair muscle and complete immunisation, or a month of decline.

Part fifteen. What the guidelines say

How do the main documents compare? On the question of whether a Kasai should come first, they agree. The 2026 American guideline recommends the Kasai as initial management with primary transplant only for decompensated disease at presentation, and the 2017 joint North American and European guideline on the cholestatic infant recommends a Kasai as soon as the diagnosis is made. European and Japanese practice is the same. On nutrition before transplant, they also agree: the 2026 guideline calls for aggressive nutritional support, and the 2017 guideline emphasises medium-chain fats and fat-soluble vitamins.

On the timing of referral there is a gap. The 2026 guideline asks for early referral at the first complication, but the 2017 cholestasis guideline does not address transplant timing, and the 2016 European liver transplant guideline is written mainly for adults. On allocation the systems differ: the United States uses PELD creatinine and MELD three point zero with exception requests, whereas Europe uses centre-based or national systems. On graft type they also differ: the 2026 guideline calls for equitable access to all technical variant grafts, while Japanese practice is dominated by living donation.

The material gap is the splenic malformation. No guideline tells us how to plan transplantation in this syndrome, when to choose a living donor over a split graft, or how to weigh donor risk against recipient benefit. All of this rests on registry data and the experience of individual centres.

Part sixteen. A stepwise plan, and the common traps

Putting this together, a pragmatic timeline for the infant whose jaundice does not clear has five steps. At three months after the Kasai, if the bilirubin is above six milligrams per decilitre or rising, refer to the transplant centre that week. Complete the evaluation by six to nine months of age, covering anatomy, heart, lungs, kidneys, nutrition and vaccines. List the child, request an exception if the PELD understates the risk, and start the living donor work-up at the same time. Choose the graft, which will usually be a living donor or split left lateral segment and rarely a whole liver. After the transplant, perform daily Doppler ultrasound in the first week, and begin planning for adherence and, eventually, transition to adult care.

Several traps recur. The first is waiting for synthetic failure: in biliary atresia the clotting time and albumin often stay normal long after the child has stopped growing. The second is trusting the weight, when ascites and organ enlargement hide muscle loss; the arm circumference is a better guide. The third is listing on the PELD score alone without an exception request. The fourth is starting the living donor work-up only after a crisis. The fifth is discovering a preduodenal portal vein or an absent vena cava in the operating room. The sixth is forgetting the heart in a child with polysplenia. The seventh is placing a gastrostomy in a child with portal hypertension, which risks varices around the stoma and peritonitis. The eighth is assuming that a child whose jaundice cleared is safe, when cirrhosis and portal hypertension can progress silently. And the ninth is treating adolescents as if they were toddlers, when adherence and transition are exactly where long-term grafts are lost.

Part seventeen. What to do differently in clinic

Let us return to the seven-month-old boy. His bilirubin was nine at three months and is now fourteen, so the Kasai has failed, and his two-year chance of survival with his own liver is in the region of one in six. He should have been referred at three months; if he has not been, he should be referred today. His PELD score may be modest because his clotting and albumin are still reasonable, but his growth failure, ascites and young age put him at real risk of dying on the list, so his team should request an exception and should begin evaluating his mother as a living donor now. A living donor left lateral segment would give him a planned date, a short cold ischaemia time and, on the national data, a graft that is more likely to last. His splenic malformation does not rule him out, but he needs an echocardiogram, venous imaging to define the portal vein and the vena cava, and an experienced centre, because complex vascular anomalies carry a first-year survival of about eighty-three percent rather than ninety-three. While he waits, he should be fed by nasogastric tube to about one hundred and thirty to one hundred and fifty percent of his energy needs, and his live vaccines should be given as early as allowed.

The broader changes for clinic are these. Write the three-month bilirubin in every letter after a Kasai, and treat a value above six as a referral trigger. Refer at the first complication rather than the third. Do not let a modest PELD score reassure you about an infant with biliary atresia. Offer families both living donor and split options. Ask for the venous anatomy in every child with the splenic malformation. And remember that the work of transplantation for biliary atresia does not end with the operation: the adolescent years, with their risk to adherence, are where the long-term outcome is decided.